VeriText AI — Semantic Similarity Analysis Report

Beyond Matching Words. Understanding Meaning.

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ine-1.0	**Embedding model**	sentence-transformers/all-MiniLM-L6-v2	**Inference device**	CPU · embed

1. Documents analysed

	Document A	Document B
File name	source_article.pdf	student_submission.pdf
Type	PDF	PDF
Words	573	670
Sentences	53 (28 compared)	56 (32 compared)
Pages	2	2

2. Summary

Coverage-weighted similarity index	**74.5%** — High overlap
Average similarity of matched sentences	90.7%
Highest single sentence similarity	100.0%
Exact textual matches	8
Near-exact textual matches	4
Semantic matches (paraphrase detected: 1)	11
Suppressed low-evidence matches	0
Matched sentences in Document A (primary evidence)	23

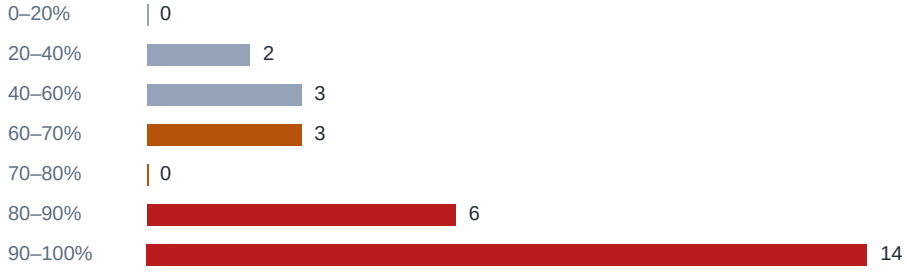
Large portions of the documents are semantically close. Check quotations and provenance before drawing conclusions.

Coverage-weighted similarity index = the best cosine score of each matched sentence in Document A (each sentence counted once, so the same evidence is never added twice) summed and divided by the number of comparable sentences in Document A. Unmatched sentences contribute zero, so the value combines how much of the document overlapped with how closely it overlapped and always lies between 0% and 100%. It is a similarity indicator, not a plagiarism percentage.

3. Match coverage

Document	Sentences compared	Sentences with a match	Sentence coverage	Word coverage
Document A	28	23	82.1%	85.3% (424/497 words)
Document B	32	21	65.6%	66.5% (403/606 words)

4. Similarity distribution (best match per sentence in Document A)



Bars show how many comparable sentences in Document A have a best-match cosine score in each range. Sentences below the flag threshold are shown so the shape of the comparison stays visible.

5. Exact textual matches (8)

Sentences that are identical after case, punctuation and whitespace normalisation.

#	Similarity	Document A (source)	Document B (comparison)	Why it was flagged
1	**100.0%** very high	§9 · p.1 In recent years, the volume of scholarly publishing has increased rapidly, and institutions have responded by deploying automated integrity screening tools.	§5 · p.1 In recent years, the volume of scholarly publishing has increased rapidly, and institutions have responded by deploying automated integrity screening tools.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).
2	**100.0%** very high	§16 · p.1 This property is precisely what paraphrase detection requires.	§9 · p.1 This property is precisely what paraphrase detection requires.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).
3	**100.0%** very high	§30 · p.1 The pipeline recovered 71% of human-labelled paraphrases at a similarity threshold of 0.62, while the character n-gram baseline recovered 18%.	§23 · p.1 The pipeline recovered 71% of human-labelled paraphrases at a similarity threshold of 0.62, while the character n-gram baseline recovered 18%.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).
4	**100.0%** very high	§31 · p.1 Exact matches were detected with equal reliability by both approaches.	§24 · p.1 Exact matches were detected with equal reliability by both approaches.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).
5	**100.0%** very high	§32 · p.1 Precision fell when the threshold was lowered to 0.55, largely because methodology boilerplate began to match across unrelated papers.	§25 · p.1 Precision fell when the threshold was lowered to 0.55, largely because methodology boilerplate began to match across unrelated papers.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).
6	**100.0%** very high	§38 · p.2 Shared definitions, standard methodological descriptions and common terminology all produce high scores legitimately.	§30 · p.1 Shared definitions, standard methodological descriptions and common terminology all produce high scores legitimately.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).
7	**100.0%** very high	§42 · p.2 Semantic similarity is a useful complement to lexical matching rather than a replacement for it.	§48 · p.2 Semantic similarity is a useful complement to lexical matching rather than a replacement for it.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).
8	**100.0%** very high	§43 · p.2 Systems that report exact, near-exact and semantic evidence separately give reviewers the information they need to distinguish copying from shared vocabulary.	§49 · p.2 Systems that report exact, near-exact and semantic evidence separately give reviewers the information they need to distinguish copying from shared vocabulary.	**Exact textual match** Identical wording after normalisation (case and punctuation ignored).

6. Near-exact textual matches (4)

Sentences whose word sequence is almost identical (minor edits, insertions or deletions).

#	Similarity	Document A (source)	Document B (comparison)	Why it was flagged
1	**99.3%** very high	§37 · p.2 A cosine similarity of 0.8 does not mean that a passage was copied; it means the two sentences occupy a similar region of the embedding space.	§29 · p.1 It is important to note that a cosine similarity of 0.8 does not mean that a passage was copied; it means the two sentences occupy a similar region of the embedding space.	**Near-exact textual match** Almost identical wording: 90% of the token sequence matches, with about 5 words changed or reordered.
2	**99.0%** very high	§22 · p.1 Prior work also documents a consistent problem: generic academic phrasing produces false positives in every embedding-based system evaluated so far.	§15 · p.1 Prior work also reports a consistent problem: generic academic phrasing produces false positives in every embedding-based system evaluated so far.	**Near-exact textual match** Almost identical wording: 95% of the token sequence matches, with about 3 words changed or reordered.

#	Similarity	Document A (source)	Document B (comparison)	Why it was flagged
3	**94.1%** very high	§27 · p.1 We exclude reference lists from matching because bibliographies repeat across papers and inflate similarity without indicating reuse.	§20 · p.1 Reference lists are excluded from matching because bibliographies repeat across papers and inflate similarity without indicating reuse.	**Near-exact textual match** Almost identical wording: 88% of the token sequence matches, with about 4 words changed or reordered.
4	**81.4%** high	§14 · p.1 A transformer encoder maps a sentence to a dense vector in a continuous space, and sentences that share meaning occupy neighbouring regions of that space.	§8 · p.1 Semantic representations provide a different route: a transformer encoder maps a sentence to a dense vector in a continuous space, and sentences that share meaning occupy neighbouring regions of that space.	**Near-exact textual match** Almost identical wording: 89% of the token sequence matches, with about 6 words changed or reordered.

7. Semantic matches (paraphrasing) (11)

Sentences with closely related meaning but different wording — the cases exact matching cannot see.

#	Similarity	Document A (source)	Document B (comparison)	Why it was flagged
1	**98.2%** very high	§21 · p.1 Studies of paraphrase identification have generally reported that contextual embeddings outperform static word vectors on semantic textual similarity benchmarks, although the margin depends heavily on domain.	§14 · p.1 Earlier studies of paraphrase identification reported that contextual embeddings tend to beat static word vectors on semantic textual similarity benchmarks, though the size of the improvement depends strongly on the domain under study.	**Semantic similarity** Sentences are about the same content (cosine 98.2%) and share 15 distinctive terms.
2	**97.7%** very high	§33 · p.1 On the annotated set, the combination of lexical and semantic signals produced the best F1 score overall.	§26 · p.1 Taken together, the combination of lexical and semantic signals produced the best F1 score on our annotated set.	**Semantic similarity** Sentences are about the same content (cosine 97.7%) and share 9 distinctive terms.
3	**93.0%** very high	§26 · p.1 Candidate pairs are then classified as exact, near-exact or semantic using lexical evidence computed independently of the encoder.	§19 · p.1 Pairs are classified as exact, near-exact or semantic based on lexical evidence that is computed independently of the encoder.	**Semantic similarity** Sentences are about the same content (cosine 93.0%) and share 10 distinctive terms.
4	**88.9%** high	§39 · p.2 Any deployment of semantic similarity for integrity screening therefore requires human review of the flagged passages, and the interface should show the matched text alongside the score so that reviewers can judge context.	§31 · p.1 Any deployment of semantic similarity for integrity screening therefore requires human review of the flagged passages.	**Semantic similarity** Sentences are about the same content (cosine 88.9%) and share 10 distinctive terms.
5	**86.1%** high	§19 · p.1 Fingerprinting methods based on hashed n-grams remain the industry standard because they are fast and easy to explain.	§12 · p.1 Fingerprinting methods that hash n-grams are still widely used because they are quick and comparatively transparent.	**Semantic similarity** Sentences are about the same content (cosine 86.1%) and share 4 distinctive terms.
6	**86.0%** high	§25 · p.1 We segment documents into sentences using an abbreviation-aware splitter, encode each sentence with a pretrained MiniLM encoder, and compare all sentence pairs by cosine similarity.	§18 · p.1 Our pipeline encodes each sentence with a pretrained MiniLM encoder after segmentation with an abbreviation-aware splitter, and then scores every pair of sentences with cosine similarity.	**Semantic similarity** Sentences are about the same content (cosine 86.0%) and share 10 distinctive terms.
7	**80.3%** high	§12 · p.1 A student who replaces half of the words in a passage can reduce a string-matching score to near zero without changing the underlying claim.	§7 · p.1 Unfortunately, a learner who substitutes half of the words within a paragraph can push a string-matching score close to zero while leaving the original claim untouched.	**Semantic similarity** Sentences are about the same content (cosine 80.3%) and share 6 distinctive terms.

#	Similarity	Document A (source)	Document B (comparison)	Why it was flagged
8	**80.1%** high	§20 · p.1 Their weakness is that any edit invalidates the hashes.	§13 · p.1 However, any edit invalidates the hashes, which limits their usefulness for rewritten text.	**Semantic similarity** Sentences are about the same content (cosine 80.1%) and share 3 distinctive terms.
9	**69.7%** moderate	§6 · p.1 We build a sentence-aligned comparison pipeline over 1,200 annotated pairs and observe that paraphrase recall improves substantially while exact-match recall remains unchanged.	§23 · p.1 The pipeline recovered 71% of human-labelled paraphrases at a similarity threshold of 0.62, while the character n-gram baseline recovered 18%.	**Semantic similarity** Meaning is close (cosine 69.7%) although the wording was rewritten — only 2 distinctive terms are shared.
10	**66.9%** moderate	§13 · p.1 Semantic representations offer an alternative.	§8 · p.1 Semantic representations provide a different route: a transformer encoder maps a sentence to a dense vector in a continuous space, and sentences that share meaning occupy neighbouring regions of that space.	**Semantic similarity** Sentences are about the same content (cosine 66.9%) and share 2 distinctive terms.
11	**64.5%** low	§10 · p.1 Most of these tools compare character n-grams or word sequences between documents.	§6 · p.1 Most of these systems rely on comparing character sequences.	**Semantic similarity** Sentences are about the same content (cosine 64.5%) and share 3 distinctive terms.

8. Paragraph-level similarities

Similarity	Document A paragraph	Document B paragraph
100.0%	¶17 Semantic similarity is a useful complement to lexical matching rather than a replacement for it. Systems that report exact, near-exact and semantic evidence separately give reviewers the information they need to distinguish copying from shared vocabulary.	¶19 Semantic similarity is a useful complement to lexical matching rather than a replacement for it. Systems that report exact, near-exact and semantic evidence separately give reviewers the information they need to distinguish copying from shared vocabulary.
99.7%	¶13 The pipeline recovered 71% of human-labelled paraphrases at a similarity threshold of 0.62, while the character n-gram baseline recovered 18%. Exact matches were detected with equal reliability by both approaches. Precision fell when the threshold was lowered to 0.55, largely because methodology bo…	¶11 The pipeline recovered 71% of human-labelled paraphrases at a similarity threshold of 0.62, while the character n-gram baseline recovered 18%. Exact matches were detected with equal reliability by both approaches. Precision fell when the threshold was lowered to 0.55, largely because methodology bo…
94.7%	¶11 We segment documents into sentences using an abbreviation-aware splitter, encode each sentence with a pretrained MiniLM encoder, and compare all sentence pairs by cosine similarity. Candidate pairs are then classified as exact, near-exact or semantic using lexical evidence computed independently of…	¶9 Our pipeline encodes each sentence with a pretrained MiniLM encoder after segmentation with an abbreviation-aware splitter, and then scores every pair of sentences with cosine similarity. Pairs are classified as exact, near-exact or semantic based on lexical evidence that is computed independently…
94.6%	¶9 Fingerprinting methods based on hashed n-grams remain the industry standard because they are fast and easy to explain. Their weakness is that any edit invalidates the hashes. Studies of paraphrase identification have generally reported that contextual embeddings outperform static word vectors on se…	¶7 Fingerprinting methods that hash n-grams are still widely used because they are quick and comparatively transparent. However, any edit invalidates the hashes, which limits their usefulness for rewritten text. Earlier studies of paraphrase identification reported that contextual embeddings tend to b…
91.3%	¶15 These results should be read carefully. A cosine similarity of 0.8 does not mean that a passage was copied; it means the two sentences occupy a similar region of the embedding space. Shared definitions, standard methodological descriptions and common terminology all produce high scores legitimately…	¶13 It is important to note that a cosine similarity of 0.8 does not mean that a passage was copied; it means the two sentences occupy a similar region of the embedding space. Shared definitions, standard methodological descriptions and common terminology all produce high scores legitimately. Any deplo…

Similarity	Document A paragraph	Document B paragraph
88.6%	¶7 Semantic representations offer an alternative. A transformer encoder maps a sentence to a dense vector in a continuous space, and sentences that share meaning occupy neighbouring regions of that space. Because the representation is learned from large corpora, it is comparatively insensitive to surf…	¶5 Semantic representations provide a different route: a transformer encoder maps a sentence to a dense vector in a continuous space, and sentences that share meaning occupy neighbouring regions of that space. This property is precisely what paraphrase detection requires.
87.7%	¶6 In recent years, the volume of scholarly publishing has increased rapidly, and institutions have responded by deploying automated integrity screening tools. Most of these tools compare character n-grams or word sequences between documents. Such approaches are effective for verbatim copying but degr…	¶4 In recent years, the volume of scholarly publishing has increased rapidly, and institutions have responded by deploying automated integrity screening tools. Most of these systems rely on comparing character sequences. Unfortunately, a learner who substitutes half of the words within a paragraph can…
69.4%	¶4 Text reuse detection has traditionally relied on string matching, which fails when authors rewrite source material rather than copying it verbatim. In this paper we examine whether sentence-level embedding models can recover reuse that lexical methods miss. We build a sentence-aligned comparison pi…	¶7 Fingerprinting methods that hash n-grams are still widely used because they are quick and comparatively transparent. However, any edit invalidates the hashes, which limits their usefulness for rewritten text. Earlier studies of paraphrase identification reported that contextual embeddings tend to b…
69.2%	¶17 Semantic similarity is a useful complement to lexical matching rather than a replacement for it. Systems that report exact, near-exact and semantic evidence separately give reviewers the information they need to distinguish copying from shared vocabulary.	¶13 It is important to note that a cosine similarity of 0.8 does not mean that a passage was copied; it means the two sentences occupy a similar region of the embedding space. Shared definitions, standard methodological descriptions and common terminology all produce high scores legitimately. Any deplo…
68.0%	¶7 Semantic representations offer an alternative. A transformer encoder maps a sentence to a dense vector in a continuous space, and sentences that share meaning occupy neighbouring regions of that space. Because the representation is learned from large corpora, it is comparatively insensitive to surf…	¶15 A word embedding represents each token as a vector of real numbers learned from co-occurrence statistics. Word2vec trains a shallow network to predict context words, while GloVe factorises a global co-occurrence matrix. Neither technique considers word order, so the resulting representation of a se…

9. Methodology

VeriText AI extracts text with PyMuPDF (PDF) and python-docx (DOCX), segments it into sentences with an abbreviation-aware rule-based splitter, and encodes every sentence with the pretrained transformer **sentence-transformers/all-MiniLM-L6-v2**. The encoder maps each sentence to a 384-dimensional vector; vectors are L2-normalised, so the cosine similarity between two sentences is their dot product. Cosine similarity is computed as (A·B)/(■A■■B■) = A·B for unit vectors.

For every comparable sentence the engine retrieves the top-k most similar sentences in the other document. Each candidate is then classified using two independent signals: (a) the embedding cosine, which captures meaning, and (b) lexical evidence — normalised string equality, token-sequence ratio and shared distinctive terms — which captures copying of words. Exact, near-exact and semantic matches are therefore reported separately. Headings, very short fragments and (by default) reference lists are excluded from matching. Sentence pairs whose only overlap is generic academic phrasing or citation context are suppressed and excluded from the headline metrics.

Thresholds in this run: semantic flag ≥ 0.62 cosine, high-similarity band ≥ 0.80, moderate band ≥ 0.65, near-exact token ratio ≥ 0.85. These are configurable operating points calibrated on a labelled evaluation set; they are not scientifically validated plagiarism boundaries.

10. Processing notes

• Document A: 9 sentences from the reference list were excluded from matching (bibliographies repeat verbatim across papers and are not evidence).
• Document B: 6 sentences from the reference list were excluded from matching (bibliographies repeat verbatim across papers and are not evidence).

11. Limitations

• The comparison covers only the two documents supplied by the user. VeriText AI does not search the web or any external corpus, so copied material from an unsupplied source cannot be detected.
• Sentence embeddings compress meaning into 384 dimensions; irony, negation subtleties, discipline-specific jargon and very short sentences can be scored inaccurately.
• Paraphrase detection is probabilistic. Legitimate similarity (shared definitions, standard method descriptions, common terminology, quotations with citation) produces high scores, and sophisticated paraphrasing can produce low scores.
• Similarity scores depend on the segmentation of both documents: different sentence boundaries can move a score up or down by several points.
• Scanned PDFs need OCR before analysis; tables, figures, footnotes and equations are extracted imperfectly.
• The similarity index is a triage aid. It is not a plagiarism percentage, and it is not evidence of intent.

12. Disclaimer

Similarity is measured, misconduct is not. VeriText AI reports semantic and lexical overlap between two documents that the user supplied. Overlap can be legitimate (quotation, common definitions, standard methodology wording, shared data description). A high score is a prompt for human review, never a finding of plagiarism.

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